

Recovering the transmission matrix of digitally modelled multimode optical fibres using PyTorch

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Abstract

This work evaluates the effectiveness of three different approaches to using PyTorch and neural networks to measure the transmission matrix (TM) of a digitally modelled step-index multimode fibre (MMF). The first two approaches attempted to find the physical parameters of a segmented MMF model but were unsuccessful due to their loss functions being too complex to navigate. The third approach was to use compressive sampling to optimise for a TM directly, which was found to be capable of producing a high fidelity TM with compression ratios as low as $\sim 20\%$ for a fibre with 189 eigenmodes. However, the high fidelity of constructed TMs did not translate to accurate image reconstruction when the direction of the model was reversed for more complex MMFs. Nonetheless, this work acts as a proof of concept that general machine learning packages like PyTorch are capable of measuring the TM of MMFs with high accuracy and speed.

1 Introduction

When fields of light propagate through opaque materials such as frosted glass or multimode optical fibres (MMFs), their spatial information becomes scrambled from the rapidly changing phases of modes that propagate the field through the medium, resulting in a characteristic speckle pattern as the field emerges from the material [1]. However, due to the linear and deterministic nature of light scattering in static media, previous studies have been able to recreate a field of light after it has propagated through the material by determining the transmission matrix (TM) of the system [2]. The TM encodes how any input field sent through an opaque material is scrambled, but its inverse also provides the ability to reverse the scrambling and regain the original field, allowing for fast and efficient information transfer with an extremely small footprint of the width of a human hair [3, 4]. This technology has numerous applications in both static and dynamic environments such as short range data transfer in a data centre [5] or minimally invasive, higher resolution imaging in micro-endoscopy [4, 6].

Despite this potential, the practical implementation of the TM for imaging through MMFs is very sensitive to its environment. Minor deformations in the MMF, or temperature fluctuations can alter how modes couple to each another, requiring the TM to be recalculated frequently [7]. Typical methods to probe the TM involve measuring the scrambling of a number of orthogonal

probe fields equal to the total number of unique modes the TM holds, but this can be too time consuming for more dynamic applications. As a result, methods have been developed to find the TM of an MMF using as few measurements as possible [8]. While these methods require using custom optimisation algorithms to find the TM, this work intends to explore the feasibility of using PyTorch, a general machine learning library, to provide a more accessible path to create neural network models that can recover the TM of a step-index MMF.

2 Theory

In this work, singularly polarised monochromatic light with a wavelength of $633nm$ was used, while under the assumption that the step-index MMF the optical field travels through is a linear medium. Due to the wavelength being much smaller than both the diameter of the fibre core, $25\mu m$, and propagation distances on the order of millimetres or centimetres, scalar diffraction theory was used to simplify the propagation of light in this work. Rather than using vectors representing the electric and magnetic fields, this method approximates a field of light using the Helmholtz equation $(\nabla^2 + k^2)E(\vec{r}) = 0$, where $k = \frac{\omega}{c}$, known as the wave number in a vacuum, ω is the frequency of the monochromatic field, c is the speed of light, and $E(\vec{r})$ is a complex-valued scalar potential dependent on the position, $\vec{r}$, in the field [9]. The solution to this approximation is a plane wave $E(\vec{r}, t) = E_0 e^{i(\omega t - k\vec{r})}$, which forms the basis for representing the field as it propagates through space [9]. The refractive index of a material is defined as $n = \sqrt{\mu_r \epsilon_r}$, with μ_r and ϵ_r representing the relative permeability and emissivity of a material, respectively. For an MMF at optical frequencies, $\mu_r \approx 1$ [9]. Therefore, the refractive index of a fibre can be represented solely in terms of the electric field, and any contribution by the magnetic component of light can be ignored in this work.

When light travels through an MMF, only linearly polarised eigenmodes of the fibre can propagate without altering their field distribution (shape), these modes are called propagation-invariant modes (PIMs) [10] and can form a basis to represent a light field when travelling through a fibre. The number of PIMs an MMF is capable of holding, and their corresponding shapes, can be found using the properties of the fibre, such as its numerical aperture, refractive index, and core diameter [11]. Fig. 1b shows a complete set of PIMs for an MMF with 21 eigenmodes. The calculated PIMs used in this work were provided to us and were originally used by Li et al. [12]. Each PIM has a constant phase velocity β_N , as it propagates through an optical fibre [10], so at the end of a fibre with length L , each PIM accumulates a global phase shift of $\beta_N L$. The superposition of these PIMs with varying global phase changes at the end of the MMF, creates the characteristic speckle pattern; the spatial information of the input field is scrambled. However, this process is linear and deterministic; for a certain shape and length of an MMF, the same input field will always return the same output field, this allows the use of a TM to represent the effects of the MMF on the input field,

$$\begin{matrix} \vec{E}_{out} \\ n^2 \times 1 \end{matrix} = \begin{matrix} \mathbf{T} \\ n^2 \times n^2 \end{matrix} \begin{matrix} \vec{E}_{in} \\ n^2 \times 1 \end{matrix} \quad (1)$$

where $\vec{E}_{in}$ and $\vec{E}_{out}$ are vectorised input and output fields, respectively, and $\mathbf{T}$ is the transmission matrix. The general shapes of each array are also given in Eq.(1), where n denotes the number of elements along the length of the input field in the real basis, before it is vectorised.

The basis in which Eq.(1) is performed is arbitrary, but two different bases were used in this work; the real and PIM bases. The real basis is discretised real space; fields in this basis

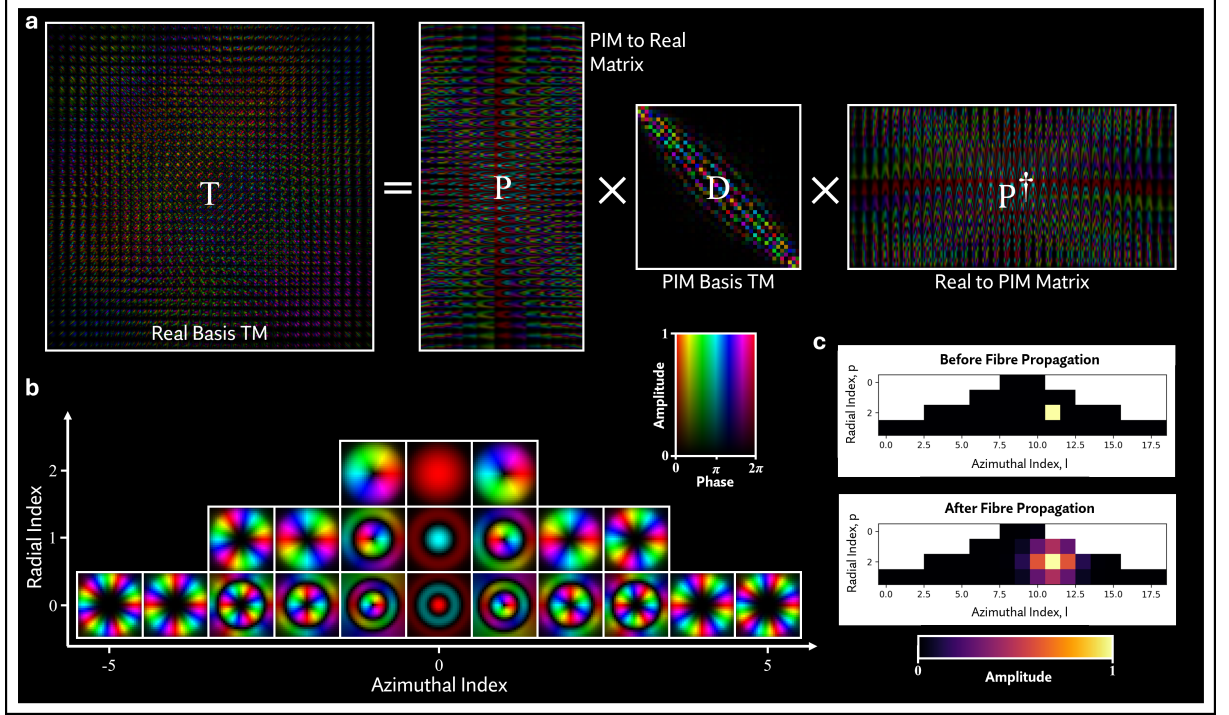


Figure 1: (a) Visual representation of changing the basis of a transmission matrix from the PIM basis (D) to the real basis (T) for a multimode optical fibre capable of holding 42 PIMs. (b) The PIM power spectrum for a multimode optical fibre holding 21 PIMs, organised by the radial and azimuthal indices of each PIM representing their radial and angular momentum respectively. The colour bar in the centre of the figure indicates the normalised amplitude and phase of plotted complex elements in (a) and (b) using brightness and colour respectively. (c) Gives an example of the expected coupling for a PIM when it propagates through an imperfect multimode optical fibre, with the relevant colour bar below the plots representing the normalised amplitude.

are expressed on a 31×31 pixel grid spanning a length of $25 \mu\text{m}$. With pixels representing the complex values of the plane waves that make up the optical field, Fig. 1b displays the PIMs using the real basis. The PIM basis expresses optical fields in terms of the eigenmodes of the MMF, as all other light is scattered outside of a fibre's core during propagation, a complete basis to represent any field that propagates through an MMF can be achieved using its available PIMs. Changing from the PIM basis to the real basis can be done using transformation matrices in the equation $T = P D P^\dagger$, where D represents the PIM basis TM and P transforms fibre mode space to real space [7, 12]. $(\cdot)^\dagger$ denotes the complex conjugate transpose of a matrix. Fig. 1a shows this transformation visually, using colour and brightness to indicate the phase and amplitude of the complex matrix elements.

While using the TM in the real basis makes finding the output field from an MMF a simple matrix-vector multiplication, the models presented in this work optimise for the TM in the PIM basis. For a perfectly straight MMF, the PIM basis TM is perfectly diagonal with each non-zero element adding the accumulated phase shift $e^{i\beta_N L}$ to a specific eigenmode over the length of the fibre. Off diagonal elements, like those seen in Fig. 1a, arise due to imperfections, such as bending, present in the MMF [10]. PIMs couple to others with similar momenta when propagating through an imperfect MMF, Fig. 1c demonstrates the expected coupling of a single PIM into others with similar indices. This coupling phenomenon translates into the off diagonal bands of power seen on the PIM basis TM.

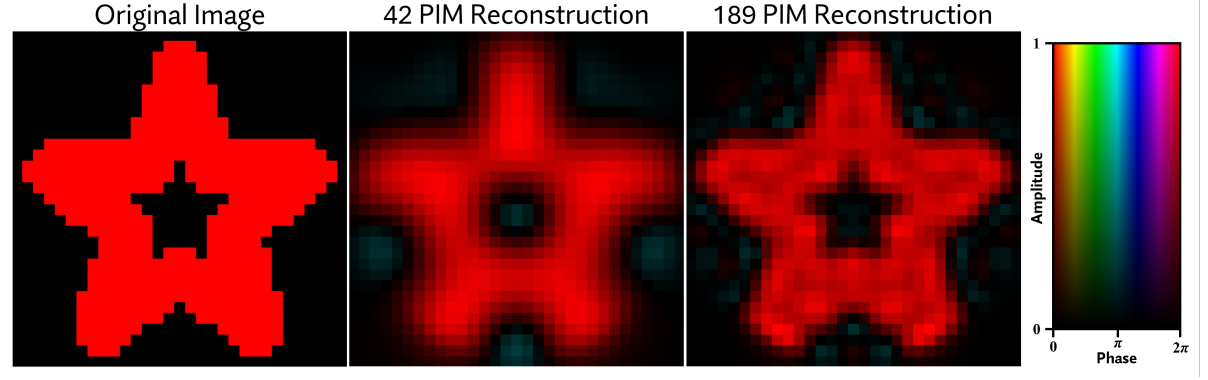


Figure 2: The best possible reconstruction of an image of a star (left plot) using a multimode fibre with 42 modes (centre plot) or 189 modes (right plot). The complex value of each pixel is represented by the colour bar using colour and brightness to represent phase and amplitude respectively.

The number of eigenmodes an MMF has is linked to the amount of information related to an input field that can fully propagate through the MMF. Because PIMs are the only modes of light capable of travelling through an MMF without losing their field profile, the output image from an MMF is a superposition of all the PIMs in their current states. This leads to the fact that some information about the input field will be lost as it enters the fibre medium and thus there is a maximum amount of information recoverable from a field that has propagated through an MMF. Fig. 2 shows the maximum possible information that can be recovered from an image entering an MMF with 42 or 189 modes. It can be seen that the MMF capable of holding more modes can have more of the original image recovered. The TM is assumed to be approximately unitary and therefore maintains the relation $\mathbf{T}^{-1} = \mathbf{T}^\dagger$ [12]. This work models MMFs capable of holding 42 or 189 PIMs, however, it is possible to model fibres with much greater numbers of eigenmodes [8].

Typically, an unknown TM can be measured by probing it with each eigenmode of the MMF and recording the corresponding output fields [8]. Physically, this can be done using a camera to measure the output field, and off-axis holography, using a coherent reference beam, to find the amplitude and phase of the field [13]. As this information about an optical field can be found in real world experiments, the digital models made in this work can have access to the same information. Fully sampling the TM involves using all N PIMs an MMF can hold, with their vectorised output fields corresponding to the n th column of the TM when represented in the PIM basis, this reconstruction method is referred to as *columnwise reconstruction* [8]. However, fully sampling a TM can be time consuming, requiring a minimum of $m = N$ individual measurements to be made. Instead of using columnwise reconstruction, the TM can be under-sampled, taking $m < N$ measurements when probing the MMF. To compensate for lost information by not fully sampling the fibre, prior knowledge of the structure of the TM in the PIM basis, such as its sparsity and strong diagonal preference, can be used to recover the majority of the lost information [8]. The *compression ratio*, $c = m/N$, quantifies how much the MMF has been sampled; any value of $c \geq 1$ can use columnwise reconstruction to recover the TM, and a value of $c < 1$ is the regime where the TM has been under-sampled.

3 Known Fibre Model

Due to experimental data from a real MMF being unavailable, a digital model was needed to simulate how light would travel through an optical fibre and act as the ground truth for the optimised models. To do this, a segmented model was created, where the MMF was split into small, perfectly straight sections, separated by phase planes which represented any bending present in that part of the fibre.

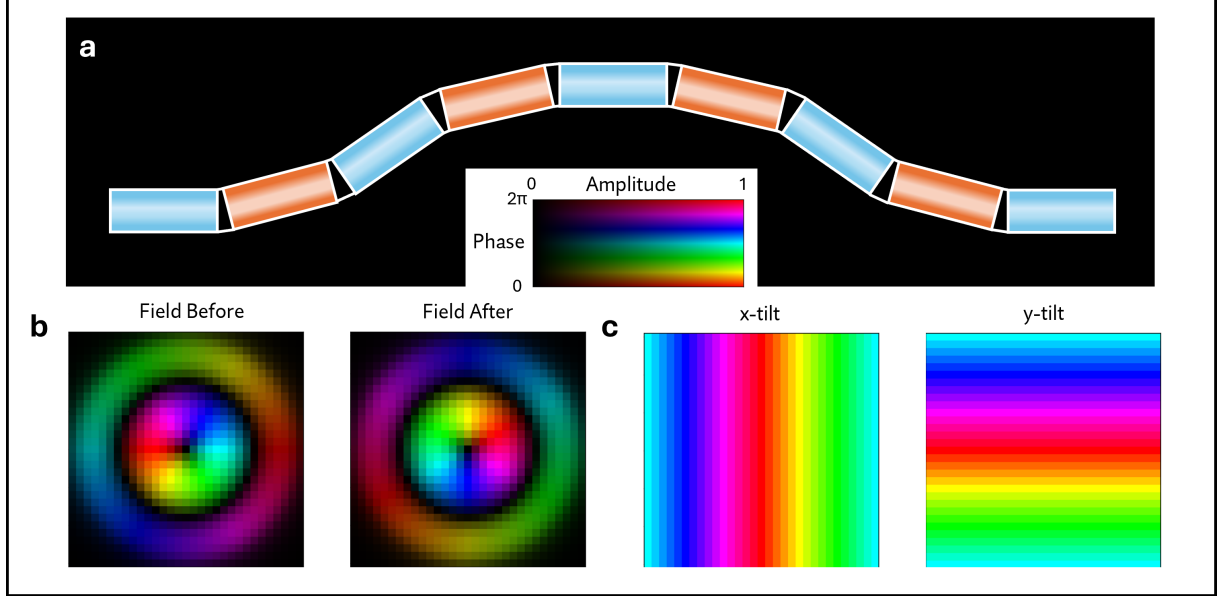


Figure 3: (a) A diagram of a multimode optical fibre split into 9 segments. (b) Shows a single PIM before and after propagating through a perfectly straight 10cm segment of MMF, keeping its overall shape while exhibiting a global phase shift of 2.62 radians. (c) Phase planes showing the phase added to a field at a certain point for a bend of 1.4509° in the x and y axes respectively.

3.1 Code Verification

Two main tests were performed to validate whether the model, and subsequent models, were working correctly.

The first check ensured that the global phase shift for each PIM was accumulating at the correct rate, while maintaining their shapes. With knowledge of the PIM's phase velocity, and the length of fibre it travels through, the expected phase shift of the PIM can be calculated by hand using the method explained in section 2. In Fig. 3b, the theoretical phase change for a mode was calculated to be 2.62 radians, when travelling through a 10cm segment of MMF, which agrees well with the simulated model.

The second check ensured that the effects of any bending in an MMF was represented correctly. To model bends in the fibre, phase planes with the second and third Zernike Polynomials encoded onto them were used to represent tilts along the x and y axes. Each plane wave that makes up the field is given a unique phase change dependent on its position relative to the x- or y-axes. From Fig. 3a, it can be seen that the area between segments is triangular. Therefore, the additional distance plane waves may travel through a bend can be approximated using trigonometry. To verify the phase planes were working correctly, the angle needed for a 2π

phase difference across a plane was found by hand. Using a wavelength of $633nm$ and the fibre's core diameter of $25\mu m$, this angle was found to be 1.4509° . When plotting a phase plane for this angle, the 2π phase difference is clear, as shown in Fig. 3c; verifying that the phase planes are working correctly.

3.2 Known Fibre Model Method

The known fibre model is used to find the ground truth TM using the following procedure:

1. **Propagation.** The inputted field propagates through a perfectly straight segment of the multimode fibre. In this scenario, the segment's TM is known to be perfectly diagonal when expressed in the PIM basis, as discussed at the end of section 2. Using the known length of the segment and the constant phase velocities for each PIM, the TM for the segment can be found. The segment TM is then transformed to the real basis for the next step.
2. **Phase Plane.** The phase plane after the segment is calculated using known angles, then the phase plane and the segment TM are multiplied together. This is done by diagonalising the resulting phase plane so that the matrices have the same shape.
3. **Repeat.** For each segment and bend present in the model, steps 1-2 are repeated, except for the final segment which does not have a phase plane afterwards.

This produces the TM in the real basis, which is used for transforming images sent through the fibre. For direct comparisons between the ground truth and the optimised models, the TM in the PIM basis is more useful. Changing from the real basis to the PIM basis requires the inverse of the operation discussed in section 2, which is $\mathbf{D} = \mathbf{P}^\dagger \mathbf{T} \mathbf{P}$.

As the ground truth model for this work, the known fibre model was used to generate training data for the future models presented in this thesis. In addition to providing the main comparison to test the effectiveness of other models.

4 Optimisation Method

Artificial neural networks (ANNs) were used for the models due to their similarity with the transmission matrix equation given in eq(1). ANNs consist of many layers of nodes, all connected with a unique weight value. An input is passed through all the layers to produce an output, with the overall weight matrix that performs this calculation acting like the transmission matrix in optics calculations, making ANNs a suitable choice for modelling optical systems.

This work utilises PyTorch, a machine learning library that provides the framework needed to make and train our models, as well as the ability to use GPUs to speed up our models [14]. For each model presented in this thesis, a python class was created to define the training parameters of the system, and perform a forward pass of the model, meaning to send any available input fields through the system to find the associated outputs in its current state. The generated output fields are then compared to the ground truth model's output fields, referred to as target fields. The mean squared error (MSE) difference between the absolute values of the fields is determined to quantify the accuracy of the training model, this is known as the loss value in machine learning.

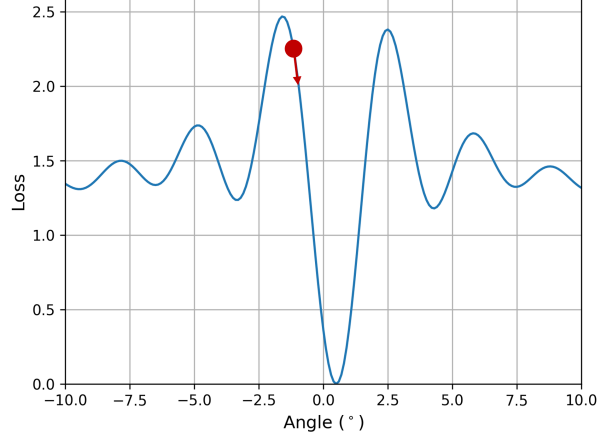


Figure 4: An example of a loss function with a single parameter, with the ground truth at 0.5° . The red dot represents how an optimiser would travel through the loss function at a certain point.

When spanned over a domain, the loss function is created. Pytorch provides optimisers that traverse the loss function, such as the one shown in Fig. 4, to find the zero loss value, which is where the model agrees with the ground truth. Optimisers typically use gradient descent to navigate the loss function, finding a minimum when the gradient becomes zero. However, due to the landscape of the loss function, it is possible for an optimiser to become stuck in a local minima; a trough in the function which is not at the zero loss value. If this happens, the model will be trained incorrectly and return incorrect results. In this work, the Adam optimiser was used to train each model. It tracks its momentum through optimisation, allowing it to avoid local minima and reach convergence faster due to less oscillations in its trajectory [15]. It also has adaptive learning rates for different parameters, avoiding slow convergence while being very computationally light [15].

Models are trained iteratively, starting with initial guesses for each parameter, the system is run and a loss value is calculated, the optimiser then tunes the parameters through a backward pass of the model to reduce the loss value. This is repeated either until convergence or until a maximum iteration limit is reached. All models presented in this work were trained using an AMD Ryzen 5 5600X CPU with 16GB of RAM.

5 Results and Discussion

The fidelity of optimised TMs can be quantified by finding their correlation between the ground truth TM and themselves,

$$\frac{1}{\|\mathbf{T}_m\| \|\mathbf{T}_k\|} \sum_{i=1}^N \sum_{j=1}^N (\mathbf{T}_m)_{ij} (\mathbf{T}_k^*)_{ij} \quad (2)$$

where i and j index the rows and columns, respectively. $(\cdot)_{ij}$ denotes the element of a matrix in row i and column j and $\|\cdot\|$ denotes the Frobenius matrix norm. $\mathbf{T}_m$ is the transmission matrix produced by the model, and $\mathbf{T}_k^*$ is the complex conjugate of the transmission matrix of the ground truth fibre. This equation produces a numerical result between 0 and 1 depending on how similar the matrices are, this value is referred to as the overlap between the matrices, and a result of 1 corresponds to the matrices being identical.

5.1 Segment Length Model

Firstly, finding the length of a perfectly straight MMF was attempted. In this scenario, the ground truth TM is perfectly diagonal when expressed in the PIM basis. Using this knowledge of the TM's shape, and the phase velocities for each of the PIMs, a TM can be constructed with an estimate of the fibre's length. This model only uses the fibre length as a parameter, and the forward pass is equivalent to eq(1).

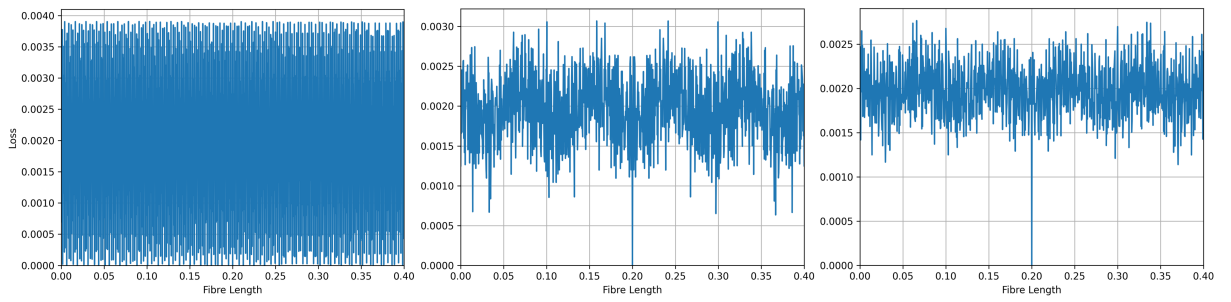


Figure 5: The loss functions for the length model, with a true fibre length of 0.2m, using different numbers of input fields. (Left) Uses 1 input field, (Centre) uses 10 input fields, and (Right) uses 42 input fields.

Varying amounts of PIMs were used as input fields to probe the TM of the MMF. Fig. 5 displays the loss functions that result from using a different number of input fields. The plot on the left shows that a single PIM does not provide enough information to accurately find the length of the fibre. This is due to the fact that PIMs only possess a 2π memory effect [12], meaning that when the fibre length increases by a distance equal to the wavelength of propagating light, a new solution is found that agrees with the ground truth model, despite being at the wrong distance. Increasing the number of input fields used, as shown in the centre plot in Fig. 5, fixes this issue due to the slightly differing phase velocities of each PIM used to probe the TM creating a large distance before overlapping on the correct solution again. Further increasing the number of input fields used to train the model makes the correct solution more defined when viewing the loss function, seen in the rightmost plot in Fig. 5. However, the noisy landscape of the loss function causes the model to fail to converge to the correct length unless an extremely precise initial guess was used, with an error of the order of magnitude equal to $0.1\mu m$. This level of knowledge about the MMF is too precise, making this model unsuitable for use in real world applications.

5.2 Segmented Bends Model

The second model presented in this work finds the bend angles between segments of an MMF, assuming perfect knowledge of the length of the fibre. This model works almost identically to the known fibre model, using the same methodology discussed in section 3.2 to find the TM. However, the bends optimiser model does not have access to the known angles of the MMF, only its length. Instead, the angles are parameters in the model, with the ground truth TM being uncovered when the correct angles are found during optimisation.

The loss function per bend is much smoother than the loss function found for the length optimiser, giving a much wider range of initial guesses where the model is able to successfully converge on the correct answer as seen in Fig. 6. At the start of each optimisation, the initial

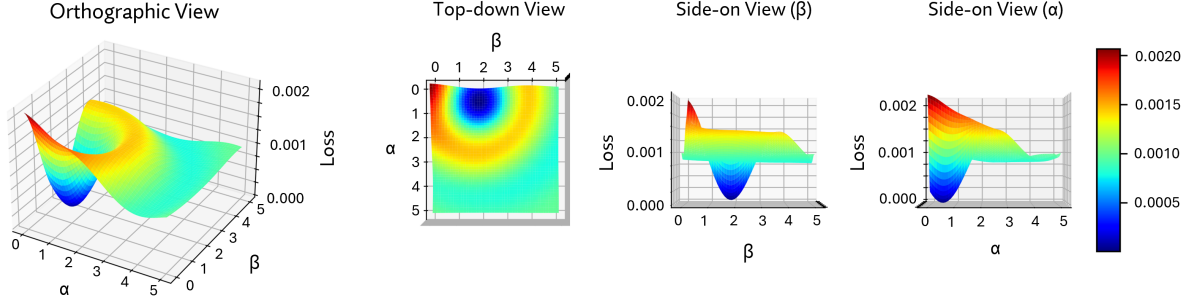


Figure 6: The loss function for a single bend in the bends optimiser model, using 5 eigenmodes of the multimode fibre as input fields, shown from multiple viewpoints. α and β represent the two angles specified in the bend with the true values being 0.5° and 1.8° respectively.

guess assumes that the fibre is perfectly straight, meaning that all angles are initially set to zero degrees before starting. Similarly to the length optimiser loss function, using more input fields smooths out the loss function more, making the ground truth easier to converge to. However, the more bends that need to be optimised for, the more complex the loss function becomes.

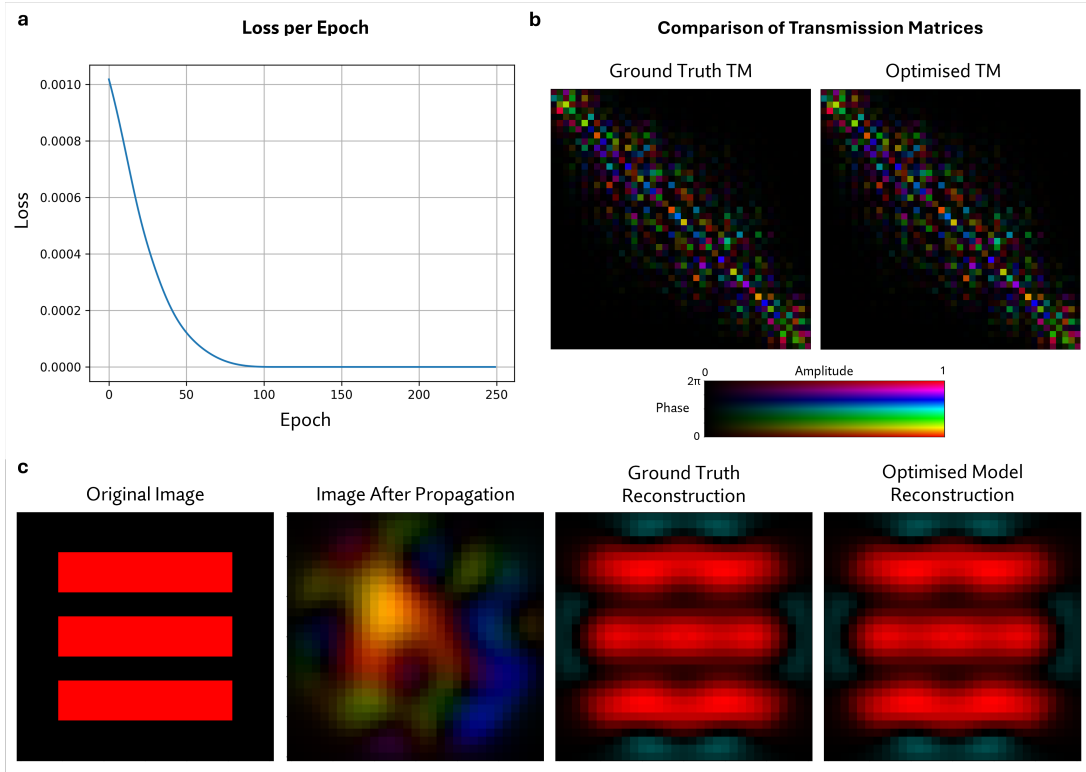


Figure 7: Training performance for a 42 mode MMF with 5 bends. (a) Loss convergence per epoch (iteration), reaching a final value of 1.69×10^{-13} . (b) Comparison between the ground truth and optimised TMs in the PIM basis. (c) Image reconstruction results: (left to right) original input field, output speckle field from the ground truth model, and reconstructions using the inverse of the ground truth and optimised TMs.

The bends optimiser model was found to consistently find the correct angles, to μ° precision, for up to five bends, which is a total of 10 unknown parameters. Fig. 7a shows that the model

converged close to the ground truth values within 100 epochs, the remaining training time was used to gain more precision on the training parameters, which took roughly 150s to train. Fig. 7b shows a visual comparison between the TM for the ground truth MMF and the TM for the optimised MMF. The overlap value between the TMs in Fig. 7 was found to be $1 - 4.2 \times 10^{-11}$. The successfully optimised model was used to reconstruct the original field sent through the MMF using the inverse of the constructed TM, as shown in Fig. 7c. The reconstructed image is nearly identical to the ground truth reconstruction, which is the best possible reconstruction of the original image. Confirmation that the optimised TM has extremely high fidelity.

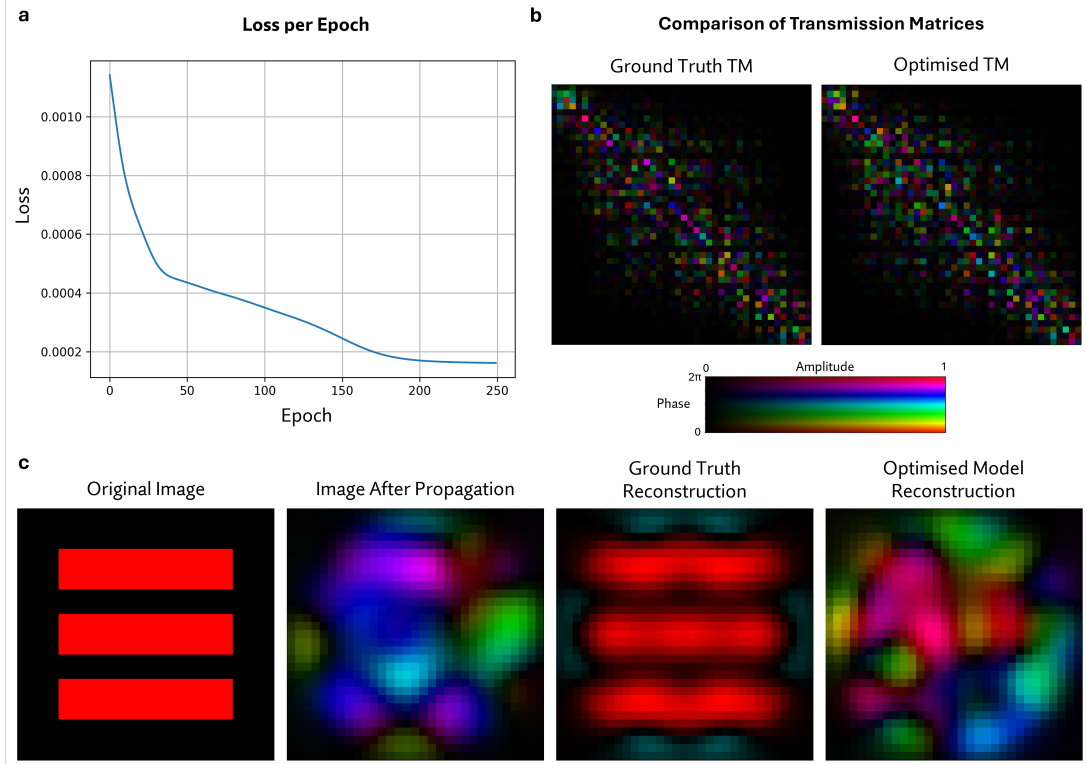


Figure 8: Training performance for a 42 mode MMF with 10 bends. (a) Loss convergence per epoch (iteration), reaching a final value of 1.62×10^{-4} . (b) Comparison between the ground truth and optimised TMs in the PIM basis. (c) Image reconstruction results: (left to right) original input field, output speckle field from the ground truth model, and reconstructions using the inverse of the ground truth and optimised TMs.

Doubling the total number of bends in the fibre model to 10, resulting in 20 unknown parameters to optimise for, causes the model to consistently fail; ending the optimisation loop in a local minima, inferred by Fig. 8a showing the loss function plateauing before reaching zero, indicating the optimiser settled in a local minima. This results in the incorrect values being calculated for the bend angles. The resulting TM, as shown in Fig. 7b, is also inaccurate because it was constructed using the wrong parameters. The result is a TM with low fidelity and an overlap value of 0.701 between itself and the ground truth TM.

Fig. 8c attempts to reconstruct an image using the poorly optimised TM, the result is a speckle field and is not recognisable from the original image. The segmented approach to modelling a real multimode optical fibre requires orders of magnitude more segments and bends than what is used in these models, creating an incredibly complex loss function for the optimiser to navigate. As this model fails to correctly converge to the correct values for simple MMFs it is

unfit for use with real world data as their applications will be heavily limited by their simplicity. The long training time required for this model also prohibits its use in dynamic applications such as micro-endoscopy, where the TM needs to be recalibrated frequently during use.

5.3 Compressive Sampling Model

The final model presented in this work uses the elements of the TM itself as the parameters to optimise for. This significantly increases the total number of parameters to find, but the forward pass of the model is reduced back to the transmission matrix equation given in Eq.(1), making a smaller ANN to train than in the bends optimiser model. The aim of this model is to under-sample the TM, using $m < N$ measurements to probe the MMF in combination with prior knowledge about the structure of the TM in the PIM basis, discussed in section 2, to compensate for the missing data that would otherwise allow the TM to be found using columnwise reconstruction.

When expressed in the PIM basis, the TM exhibits a strong preference for power along the main diagonal of the matrix, with off-diagonal elements occurring in bands with diminishing power, as discussed in section 2. The expected coupling between the n th PIM and the other modes was modelled using a 2D Gaussian distribution within the bounds of the PIM power spectrum,

$$f(l, p) = \exp \left[-\frac{(l - l_0)^2}{2\sigma_l^2} - \frac{(p - p_0)^2}{2\sigma_p^2} \right] \quad (3)$$

l_0 and p_0 are the indices corresponding to the n th PIM in the PIM power spectrum. l and p are the indices of a coupled PIM, and σ_l and σ_p are standard deviations to represent the estimated coupling strength. A matrix can be built to estimate the structure of the TM in the PIM basis by repeating Eq.(3) for each PIM and each possible coupling pair, where the results for the n th PIM corresponds to the n th column of the matrix, which will be referred to as the *estimated support matrix*. The way in which elements of the PIM power spectrum are reordered to produce the n th column of the estimated support matrix is arbitrary, but this work follows the ordering convention set in references [8, 12].

The models based on the segmented view of an MMF propagate fields by applying phase changes to the superposition of PIMs that make up the field in the fibre. Because they interact with the PIMs directly, there is no necessary prior knowledge needed to guide the optimiser to the correct TM form, as the quasi-diagonal structure is a result of the behaviour of PIMs. The forward pass of the compressive sampling model does not involve the PIMs of the MMF, so the correct structure of the TM is instead obtained using the estimated support vector as a guide. To do this, a new term is added to the loss function L ,

$$L = \frac{1}{m} \sum_{k=1}^m |Y_k - \hat{Y}_k|^2 + \frac{\lambda}{N^2} \sum_{i=1}^N \sum_{j=1}^N |((\mathbf{1} - \mathbf{W}_{ij})\mathbf{T}_{ij})|^2 \quad (4)$$

The first term is the mean squared error between the output fields (Y_k) and the target fields ($\hat{Y}_k$), and m is the number of independent measurements used. The second term is a regularisation term used to enforce the quasi-diagonal structure of the TM. N represents the number of PIMs the MMF has, $\mathbf{1}$ is an $N \times N$ matrix of ones, $\mathbf{W}$ is the estimated support matrix and $\mathbf{T}$ is the transmission matrix. The subscripts i and j index the rows and columns of a matrix, respectively, and λ is a scalar number that represents the relative importance of the regularisation term compared to the MSE term, a range of values was manually tested before $\lambda = 0.01$ was chosen for an

MMF with 42 PIMs, and $\lambda = 0.25$ was chosen for an MMF with 189 PIMs. Apart from an altered loss function, the optimisation method is the same as the method specified in section 4.

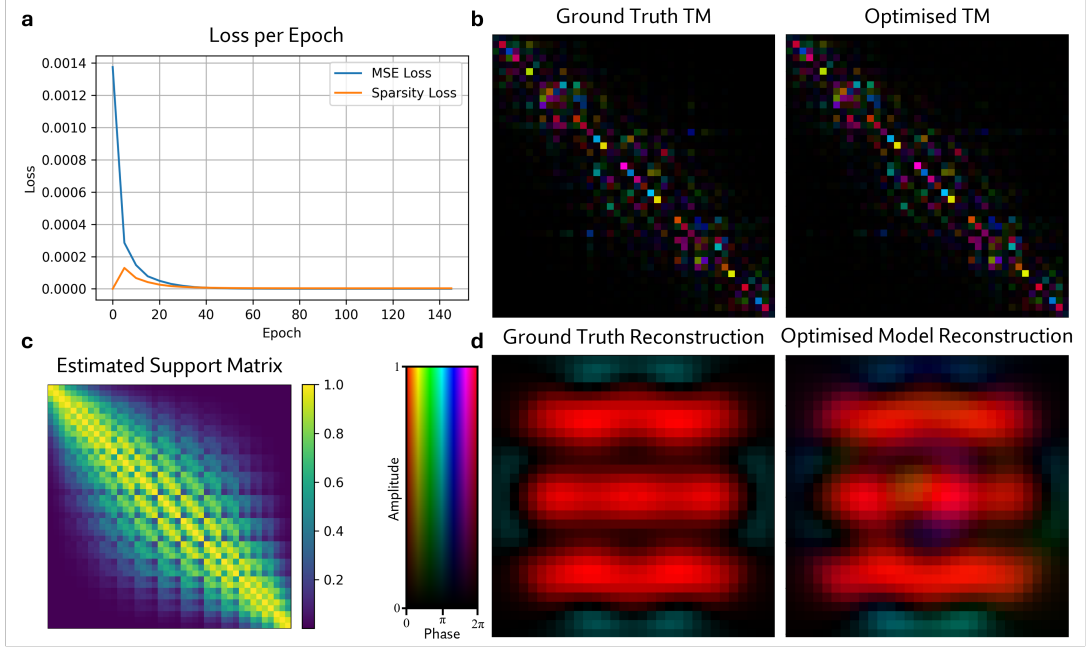


Figure 9: Results related to the compressive sampling model for a multimode fibre with 42 eigenmodes and 10 bends when fully sampled. **(a)** A graph of the two terms in the loss function per epoch (iteration) of the optimisation process, ending with a final loss sum of 3.15×10^{-6} . **(b)** Visualises the ground truth TM and the TM found using the model. **(c)** Visualises the estimated support matrix used during training using standard deviations $\sigma_l = 3$ and $\sigma_p = 2$. **(d)** Shows the reconstruction of an image using either the inverse of the ground truth TM or the inverse of the optimised model TM.

The model, in the fully sampled case, is successful in finding the ground truth TM to a very high fidelity, with an overlap value of 0.987 and took roughly 0.1s to complete the training process. Fig. 9a shows that the model finished training in 150 steps and found good agreement with the ground truth model after only 40 epochs. The high fidelity optimised TM, shown in Fig. 9b, very closely resembles the ground truth TM and its inverse reconstructs an image that was sent through the fibre model to a high degree, as seen in Fig. 9d. For MMFs that support 42 PIMs, an estimated support matrix with standard deviations $\sigma_l = 3$ and $\sigma_p = 2$ was found to work well during the optimisation process, when visualised in Fig. 9c the chosen values overestimate the support of the ground truth TM. This was done to allow many different TM structures to be optimised for, while only requiring one support matrix to enforce the prior knowledge of the structure of the TM in the PIM basis.

An MMF with 189 eigenmodes was also computed using the compressive sampling model with the aim of achieving a higher reconstruction fidelity of the original image passed through the fibre model, not just with the best possible reconstruction using only 42 PIMs.

Despite a larger TM needing to be optimised and more training data needed for training, the optimisation took roughly 0.4s to compute when fully sampled. Fig. 10a shows that the optimisation process was able to run completely within the same number of epochs as with the 42 PIM model and generated a high fidelity TM, seen in Fig. 10b, with an overlap value of 0.988. With more PIMs to store more information about the original image sent through the fibre, a much better ground truth reconstruction of the image can be achieved, as seen in Fig. 10d. The

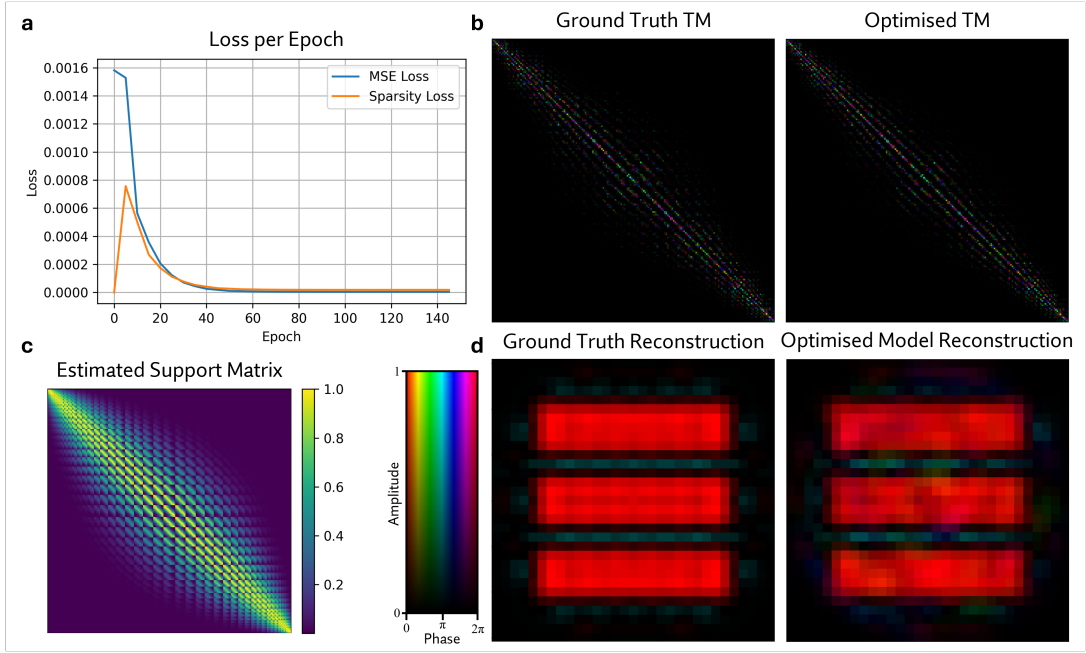


Figure 10: Results related to the compressive sampling model for a multimode fibre with 189 eigenmodes and 10 bends when fully sampled. **(a)** A graph of the two terms in the loss function per epoch (iteration) of the optimisation process, ending with a final loss sum of 2.28×10^{-5} . **(b)** Visualises the ground truth TM and the TM found using the model. **(c)** Visualises the estimated support matrix used during training using standard deviations $\sigma_l = 4$ and $\sigma_p = 2$. **(d)** Shows the reconstruction of an image using either the inverse of the ground truth TM or the inverse of the optimised model TM.

reconstruction using the inverse of the optimised TM is also of a high quality, with the three horizontal bars visually resolved. Again, an overestimate for the estimated support matrix was used during the training process, here standard deviations of $\sigma_l = 4$ and $\sigma_p = 2$ were found to work well with 189 mode fibres.

Fig. 11 plots the mean correlation between under-sampled TMs and the ground truth TM as a function of the compression ratio, c . For both MMFs with 42 or 189 PIMs, the compressive sampling model achieves a high fidelity much quicker than using columnwise reconstruction which scales linearly with the number of measurements taken. The 189 PIM fibre reaches an overlap value of 0.9 at $c \approx 0.2$, which is much faster than the 42 PIM fibre that requires $c \approx 0.5$.

How different compression ratios affect the model's ability to reconstruct an image is shown with an example in Fig. 12 where the University of Exeter logo is reconstructed using the TMs generated with different compression ratios of training data for an MMF with 189 PIMs. Despite a compression ratio of roughly 0.2 correlating to an overlap value of 0.9, this is not a strong enough correlation with the ground truth TM in order for the reconstructed image to resemble the original image in any way. Once $c = 0.4$, corresponding to an overlap value of ~ 0.95 , the reconstructed image is discernible as the University of Exeter logo and any images reconstructed using a compression ratio $c \geq 0.6$, corresponding to overlap values ≥ 0.98 , do not have any visually resolvable differences.

The compressive sampling model is the most promising of the three models presented in this work. The model is very robust; changes in the estimated support matrix do not largely affect the model's ability to converge to a high fidelity TM and it does not require an accurate initial guess for the length or bend angles of the MMF. Fig. 11 shows that the model works

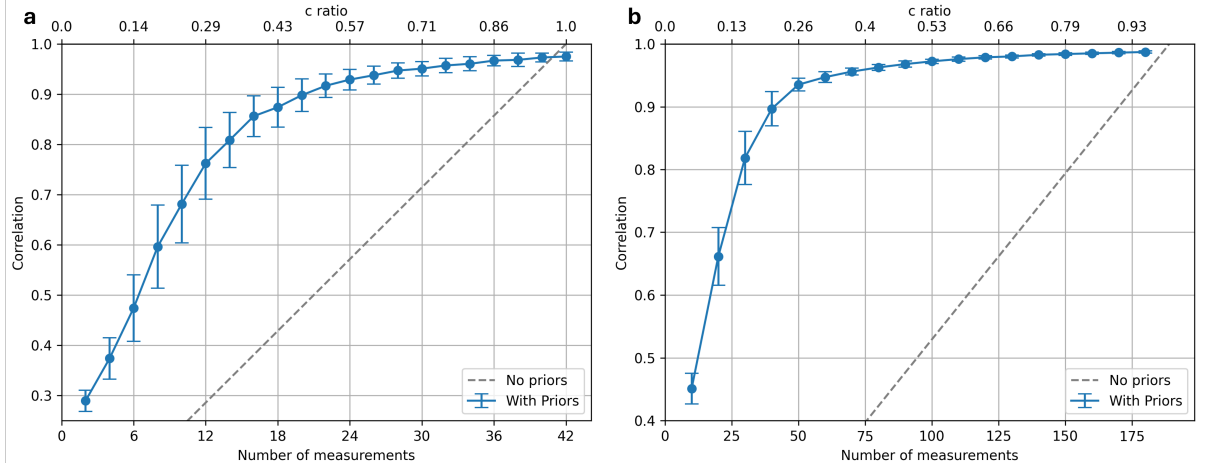


Figure 11: Graphs showing the correlation between the ground truth and optimised TMs versus the compression ratio, c , or number of measurements, m , used during the training of the compression ratio model (with priors) and using columnwise reconstruction (no priors). Each data point represents the mean of 100 separately run models computing to find the TM of the same MMF, but using different input fields, with the error bars displaying the standard deviation from the mean. **(a)** Shows the relation for an MMF with 42 PIMs. **(b)** Shows the relation for an MMF with 189 PIMs.

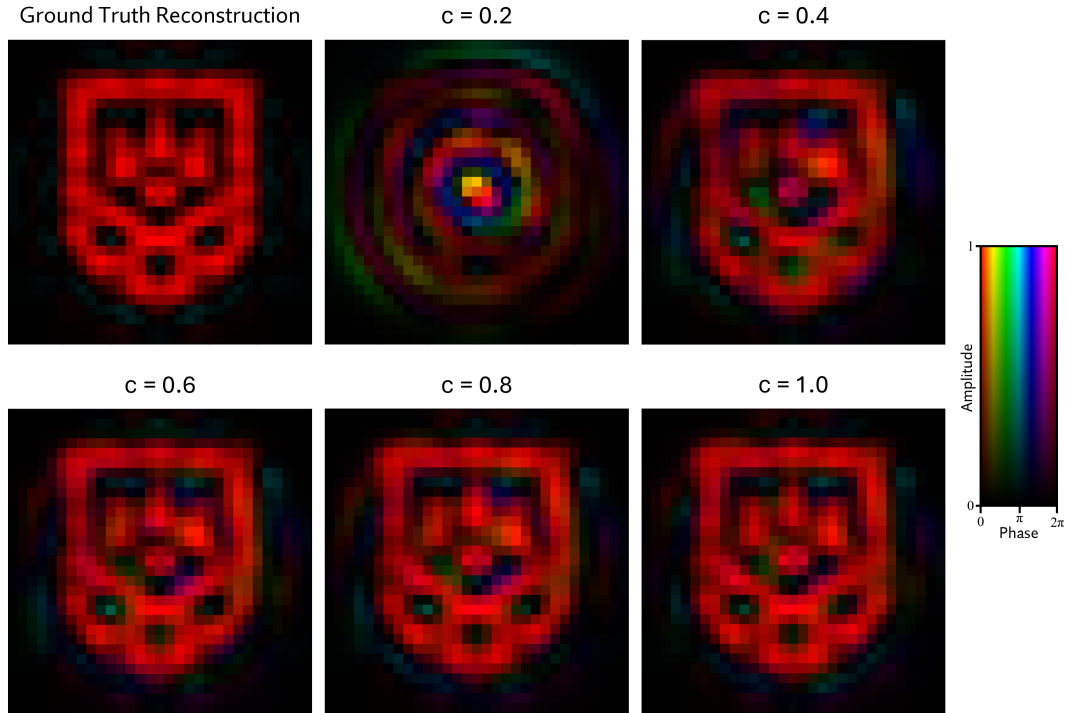


Figure 12: Reconstructions of the University of Exeter logo for a range of different compressive ratios. Modelled using a 189 PIM multimode fibre with 10 bends.

consistently to find a TM that has a high correlation with the ground truth model while in the under-sampled regime. Additionally, Fig. 12 shows that a model with a compression ratio of $c = 0.6$ can reproduce a visually accurate image while only probing the MMF with 60% of the theoretical maximum number of measurements required to fully sample the TM. The model

itself also trains quickly, making it a good choice for dynamic MMF applications where the TM needs to be recalculated frequently.

However, the more bends present in the segmented model used to generate the ground truth data for training the compressive sampling model, the less a reconstructed image using the inverse of the optimised TM resembles the original image. This is despite the optimised TM having a strong correlation with the ground truth TM, as shown in Fig. 13.

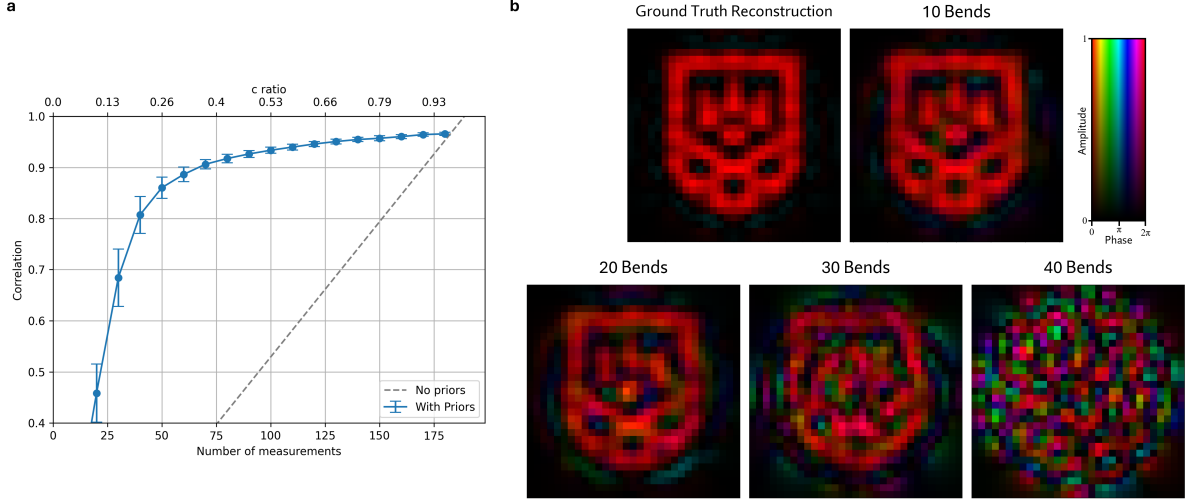


Figure 13: (a) Graph showing the correlation between the optimised TM and the ground truth TM, for a fibre with 40 bends, across a range of different compression ratios. Each data point represents the mean overlap value of 100 models ran at a specific compression ratio, with the error bars displaying the standard deviation of the data. (b) Reconstructions of the University of Exeter logo for a fully sampled model where the ground truth fibre has a differing number of bends.

The high sensitivity of the TM inversion to small numerical discrepancies was unexpected. While previous work by Li et al. [8] demonstrated a correlation of ~ 0.6 was a high enough fidelity to recreate an image that passed through an MMF, the TMs in this work required much higher correlations ~ 0.95 to reproduce similar results. This discrepancy likely stems from the study using a larger fibre with 754 PIMs and a higher spatial resolution, providing their model with more information to train from and consequently a better measurement of the TM. Fig. 11 shows the correlation depending on compression ratio increases faster for the 189 PIM fibre than its 42 PIM counterpart, suggesting that MMFs with larger numbers of supported PIMs require lower compression ratios to recover their TMs to a high fidelity.

Furthermore, while the compressive sampling model works well to find a matrix that can effectively map an input field to the correct output, it does not know that the TM needs to be unitary in order to represent the physical reverse of the system. This is evidenced by the drop in overlap value when the TMs are inverted, where small training errors become amplified and significantly impact the reconstruction accuracy.

6 Conclusion

This work evaluated the effectiveness of three different approaches to using PyTorch to measure the transmission matrix of a step-index multimode fibre. While the models based on finding the

physical parameters of a segmented MMF struggled to correctly converge to the ground truth values due to complex loss functions, the compressive sampling model proved itself to be much more robust; converging to a high fidelity TM without requiring precise knowledge of the length or geometry of the MMF. The model made accurate recreations of input fields for a compression ratio as low as $c = 0.6$ for a fibre capable of holding 189 PIMs, with a fast training time of $0.4s$, suggesting the model is suitable for real-time applications, such as in micro-endoscopy, where the TM needs to be frequently recalibrated due to movement deforming the MMF.

However, a discrepancy between the accuracy of the forward pass of the model, given by high correlation between the ground truth and optimised model TMs did not translate to accurate image reconstruction when the direction of the model was reversed for more complex MMFs. A small amount of eigenmodes of the fibre, low field resolution, and no unitary constraint in the compressive sampling model are expected to be why the model starts to break down. As when the number of fibre segments increases, small training errors compound and have a large impact on the inverse TM.

Future work in this area should focus on using higher resolution data while adding a unitary constraint term to the loss function; enforcing $\mathbf{T}^{-1} = \mathbf{T}^\dagger$, and allowing the system to be physically reversed for more complex MMFs. Overall, this work demonstrates that a general machine learning library such as PyTorch is capable of being used to measure the transmission matrix of a step-index multimode fibre with high accuracy in a timely fashion.

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